

$$C=1+2+3+4+5+\dots = -1/12$$



Srinivasa Ramanujan

- Indian mathematician who, despite a lack of formal training, made profound contributions to complex analysis, number theory, infinite series, and continued fractions.
- unique mathematical insights, often presented without formal proof
- famous after corresponding with G. H. Hardy at Cambridge University, got hella breakthroughs
- died of hepatic amoebiasis in 1920



First let's prove two other less absurd claims

$$A = 1 - 1 + 1 - 1 + 1 - 1 + \dots = -1/12$$

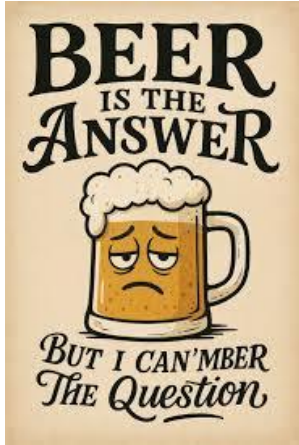
$$B = 1 - 2 + 3 - 4 + 5 - 6 + \dots = 1/4$$



Going back to the original claim

$$A=1-1+1-1+1-1+\dots = \frac{1}{2}$$

$$B=1-2+3-4+5-6+\dots = \frac{1}{4}$$



$$C=1+2+3+4+5+\dots = -\frac{1}{12}$$



$$C=1+2+3+4+5+\dots = -1/12$$



Whats wrong

-The steps we just took are not valid under **standard calculus**.



-Series A, B, and C are all **divergent**. They do not tend to a single number, but grow infinitely large (or oscillate infinitely).

-Standard algebra allows **commutativity** and **associativity** (rearranging and grouping terms) *only* if the original series is **convergent**

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-For divergent series, rearranging terms can lead to *any* result, making the algebraic proof invalid in the traditional sense.

Ramanujan Summations

The Fix: The value $C = -1/12$ is not the traditional sum, but a value **assigned** to the series using advanced methods.

Cesàro Summations: A technique that assigns values to some infinite sums that do not converge conventionally.

- **Definition:** The Cesàro sum is the limit of the sequence of arithmetic means of the partial sums.

Countable Infinity: The initial problem is resolved by adopting a broader definition of summation that works within the system of countable infinity.



Freaky Ramanujan Summation used in:

Bosonic String Theory:

The Casimir Effect (Quantum Physics):

